Conflict and Peace

Week 13: Fiscal capacity of the state

Seung-hun Lee

Taipei School of Economics
National Tsing Hua University

November 28th, 2025

Development of fiscal capacity is a key feature of development

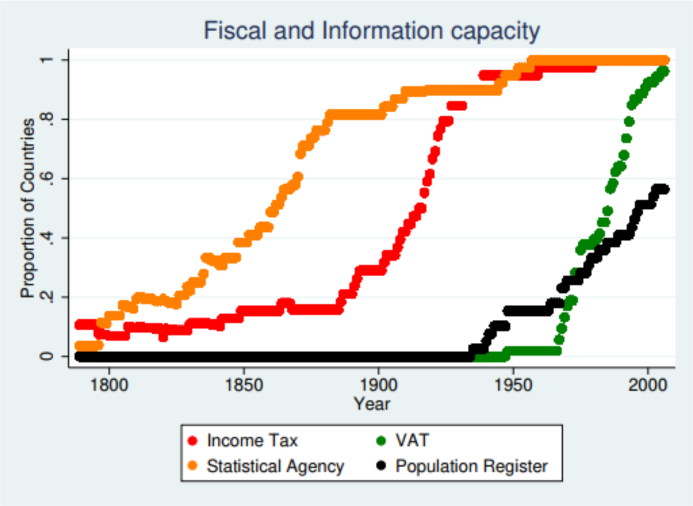
Expansion of government size and fiscal capacity investment in the last century

- Tax share of GDP: 10% in 1910 $\rightarrow$ 30-50% today
- Tax capacity improved following implementation of broad-based taxes (income tax)
- However, this explains the trend only in developed countries
  - So economic growth $\rightarrow$ better fiscal capacity? Not regularly
  - Separate investment in capacities are key

But in poor states, no noticeable increase in fiscal capacities

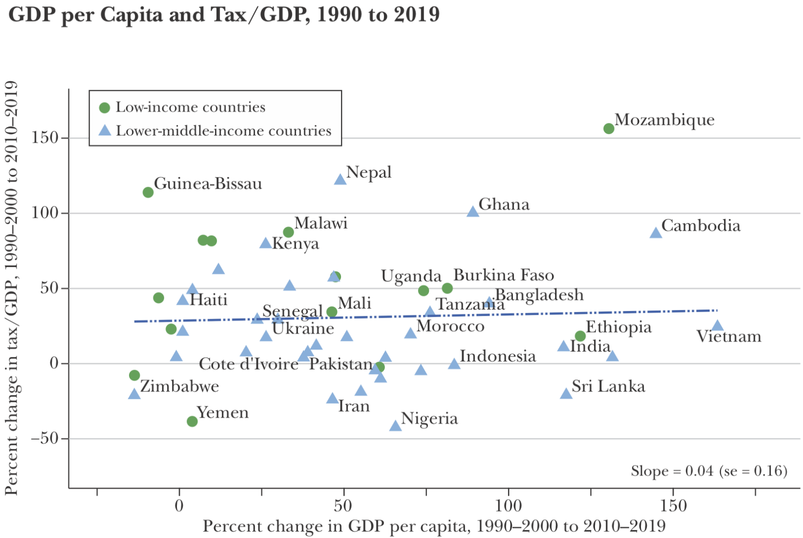
- Raise significantly less revenues than developed countries
- Lag behind in the implementation of modern tax technologies
  - Property registers are manually updated, if updated at all
  - Do not utilize third-party information that ensures accurate tax information
  - Less digitalized systems, more loopholes in general

Rise of informational capacity and taxation over time



Fiscal and Information Capacity in a sample of 39 countries

Economic growth does not automatically lead to increased tax



Source: UNU-WIDER (2022a, b).

Less capable states rely more on distortionary and narrow based taxes

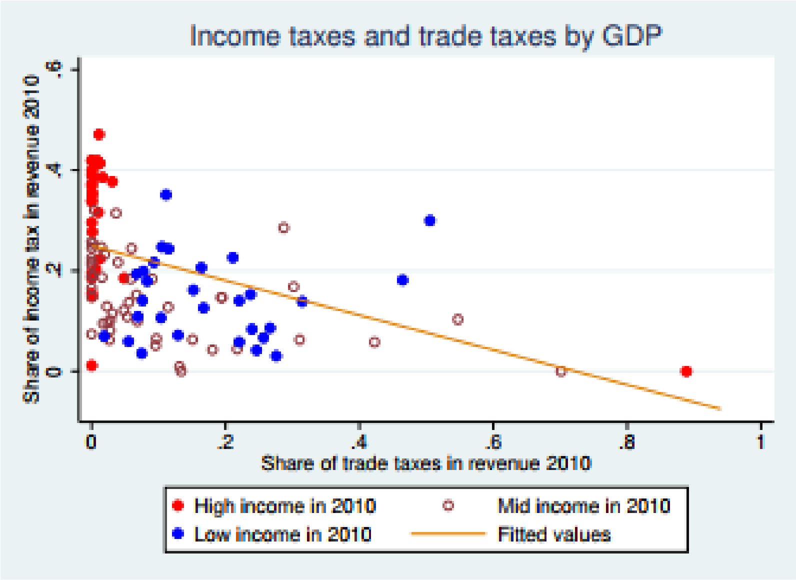
Why do they depend on such taxation

- Broad-base taxes require registries that cover formal sector, and audits
- Narrow-base taxes rely on easily observable bases (trade ports, mines etc)
- Large informal economy and underdeveloped finance sector: Opens up loopholes
- Extractive for select groups by shifting tax burden onto them
- Bad equilibrium: Fiscal capacity$\Downarrow \rightarrow$ Public goods and economic development $\Downarrow$

What the data tells us

- High income countries rely on broad-based taxes
- Narrow base taxes (tariffs) more observable in low-capacity countries

High income → Income tax, Lower income → Trade tax



Roadmap for today

1. Overview of the topic

2. Tax structure across developing countries
   2.1 Gordon, Li (2009): Tax structures in developing countries: Many puzzles and a possible explanation
   2.2 Okunogbe, Tourek (2024): How can lower-income countries collect more taxes? The role of technology, tax agents, and politics
   2.3 Jensen (2022): Employment Structure and the Rise of the Modern Tax System
   2.4 Olsson, Baaz, and Martinsson (2020): Fiscal capacity in post-conflict states: Evidence from trade on Congo River [student presentation]

3. Barriers to taxation in developing countries
   3.1 Balán, Bergeron, Tourek, Weigel (2022): Local Elites as State Capacity: How City Chiefs Use Local Information to Increase Tax Compliance in the DRC
   3.2 Henn, Mugaruka, Ortiz, Sánchez de la Sierra, Wu (2025): Monopoly of Taxation without a Monopoly of Violence: The Weak State's Trade-off from Taxation
   3.3 Ch, Shapiro, Steel, Vargas (2018): Endogenous Taxation in Ongoing Internal Conflict: The Case of Colombia [student presentation]

Gordon, Li (2009):
Tax structures in developing countries: Many puzzles and a possible explanation

Q: Do presence of informal sectors affect choice of tax policies?

Major difference in tax structures across developing and developed countries

- Some could be attributed to social preferences over public goods and redistribution
- But many differences with predicted optimal taxation in generic settings
  - These taxation preserves production efficiency under plausible assumptions
  - Rule out tariffs, capital income taxes, differential consumption tax, inflation tax
- Drastic diffrences in developing countries
  - Less personal income tax, more tariffs, corporate income tax, and seignorage

This paper proposes that existence of informal sector explains different tax structures

- Incorporates enforcement problems in developing countries due to informal economy
  - Informal economies: Use cash over financial intermediaries
  - Taxes on formal economy shifts activities to informal economy $\rightarrow$ Optimal tax rate $\Downarrow$
  - Justifies inflation tax, differential tax across industries, and tariffs

Summary of the difference in tax choices across country income

Table 1
Sources of government revenue (1996–2001).

GDP per capita	Tax revenue (% of GDP)	Income taxes (% of revenue)	Corporate income tax (% of income taxes)	Consumption and production taxes (% of revenue)	Border taxes (% of revenue)	Inflation rate	Seignorage income (% of revenue)	Informal economy (% of GDP)
<$745	14.1	35.9	53.7	43.5	16.4	10.6	21.8	26.4
$746–2975	16.7	31.5	49.1	51.8	9.3	15.7	24.9	29.5
$2976–9205	20.2	29.4	30.3	53.1	5.4	7.4	6.0	32.5
All developing	17.6	31.2	42.3	51.2	8.6	11.8	16.3	30.1
>$9,206	25.0	54.3	17.8	32.9	0.7	2.2	1.7	14.0

Notes: Authors' calculations based on available data between 1996 and 2001 from Government Finance Statistics (IMF, 2004a), International Finance Statistics (IMF, 2004b), and World Development Indicators (World Bank, 2003). The ranges for GDP per capita follow the World Bank 2003 classification of low income, lower middle income, middle income and high income countries. Seignorage is measured as the increase in reserve money and currency in circulation. Estimates of the size of the informal economy in 1999 in Column (9) are from Friedrich Schneider (2002), who uses the currency demand approach in estimation. Data within each cell are weighed averages. Tax revenue (% of GDP), inflation rate, and the size of the informal economy (% of GDP) are weighted by GDP of each country. Corporate income tax (% of income taxes) is weighed by the total income tax revenue of each country. All other data are weighted by the tax revenue of each country.

- High corporate income tax, tariffs, seignorage in developing countries
- Less personal income tax
- Statutory tax rates are similar across different countries!

Tax policy with limited information: Differential excise taxes

Characterizing informal economy into a model

- Informal economy not observed by government
  - Thus, taxes cannot be levied to such firms
  - So profits for such firms are revenues - wage cost - capital costs
- Formal firms use financial systems and have higher outputs, but subject to tax
- Firms become formal if net profits exceed those from informal economy
  - High excise taxes reduce the net profits $\rightarrow$ tax revenue drops

Variation across industries

- Sectors with low increase in output from formalizing $\rightarrow$ lower tax rates than desired
- This results in distortion in domestic production, which are offset by tariffs on imports
- Inflation tax and red tape on lightly taxed sectors may be attractive

How to increase benefits of using financial sector

Determinants of benefits to using financial sectors

- Capital intensive firms: They need lending to finance purchase of capital
- If firms operate at wide distance, financial sector reduces the cost
- High revenue per employee: More cash transaction exposes to internal theft
  - Service sectors tend to be less capital intensive and easily transfer to informal economy
- Firms that do not formalize avoid corporate and excise tax, but pay inflation tax
- Thus, taxes are kept low for non capital-intensive sectors
  - Capital-intensive ones more likely to stick to formal even without intervention
  - This is meant to shift informal economies to formal ones
- This explains high capital income tax in developing countries (vs. income tax)
- Inflation taxes also raise cost of keeping cash, inducing firms to formalize

Motivating uses of other anti-competitive measures

Tariffs

- Higher tariff protection for capital intensive sectors
  - Meant to offset distortions due to uneven domestic tax enforcement
  - Specifically, aim is to induce production to a sector more likely to formalize

Other anti-competitive measures

- Provide monopoly power to capital-intensive firms
- Impose tighter regulation or monitoring on informal firms
  - Even if revenue from such firms cannot be observed, other activities could be
  - Higher red tape measure on those firms as monitoring device

Key takeaways

Existance of informal firms twist the choice of optimal taxation

- More capital taxation, tariffs, and inflation taxes
- This is determined by the returns to using financial sectors
  - If financial sectors provide benefits, less incentives to join informal economy
  - As developing countries have less developed finance sectors, high informality
- Thus, models should incorporate possiblity of informal economies

However, leaving informal economies as is is undesirable

- Weakens capacity to provide essential public goods
- Resource misallocation$\Uparrow$ and innovation$\Downarrow$ (cost of hiding leave little for innovation)
- Thus, efforts to induce firms to formalize through third-party information, legal measures, and expansion of electronic filing (Benhassine et al 2018)

Okunogbe, Tourek (2024): How can lower-income countries collect more taxes? The role of technology, tax agents, and politics

Q: How do you improve tax system in developing countries?

Increasing tax revenues is a major goal in many low/middle income countries

- Generally have lower share of their GDP as tax revenues
- Low taxation is a symptom of economic and institutional constraints
  - Tax base not large enough, lack of constraint on redistribution/taxes
- Does economic growth automatically expand tax capacity? Not exactly
  - Separate investments in improving the tax system required

This paper reviews recent developments in tax administration in developing countries

- Identifying taxpayers (expanding tax registries and databases)
- Detecting tax burden (usage of third-party information)
- Collect taxes to state coffers (Billing systems, legal measures for noncompliance)

Technology in tax collection

Identification: Cost of linking identity info across government agencies and registries⇓

- National ID (Ghana Card), Digital mapping of properties (Liberia)
- ID information may need to be combined with other measures (Okunogbe 2023)

Detection: Collect third-party info (declaration of tax liabilities by third person)

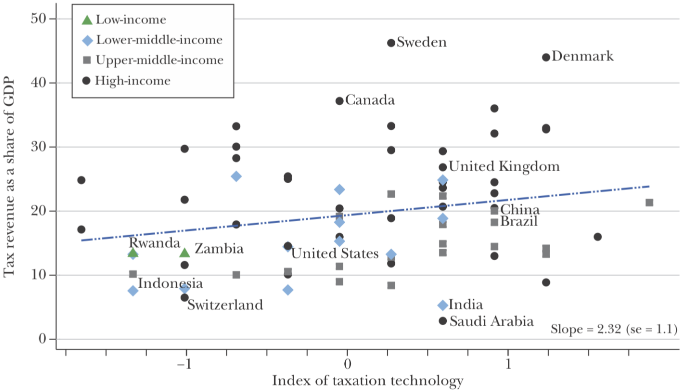
- ‘Digital trails’ from value-added taxes and employment taxes
- Requires countries to cross-check information using multiple sources
- Need to close international loopholes

Collection: Increases in electronic filing and mobile payment options

- Reduces cost of monitoring and Compliance
- May disadvantage those less familiar with electronics
  (rural, less-educated, less-established firms)

Cross-country evidence on technology and collection

Taxation Technology and Tax-to-GDP



Source: Tax revenues as a share of GDP measure drawn from the International Survey on Revenue Administration (CIAT, IMF, IOTA, and OECD 2022), using values for 2018. Taxation technology measure drawn from OECD (2023) Global Survey on Digitalisation.

Role of tax collectors

What determines the effectiveness of officials

- Deployment: low citizen-to-staff ratio and appropriate allocation
  - Low burden per staff
  - Specialization: Matching staff depending on tax types and taxpayer traits
- Incentives: Tie staff revenues to tax revenue generations
  - Risk: Could increase scope for corruption
  - Monitoring staff with audits crucial

Interaction with technology

- Substitute? Obviate need for in-person Interaction
- Complement? Help expedite tedious processes and collecting contextual knowledge
- Balancing the two forces is key (and unresolved question)
- Resistance to adopting the technological change

Political incentives for taxation

How do political incentives determine tax collection

- Tax$\Uparrow$ $\rightarrow$ fund public services, social programs, infrastructure $\rightarrow$ support re-election
- Tax$\Uparrow$ $\rightarrow$ Citizen demand, expectation for accountability $\Uparrow$ $\rightarrow$ not help that much

What are the factors shaping success of tax collection?

- Alternative revenue: Dampen motivation for Taxation
- Political competition: Competition for better governance (through public services)
- Availability of tech: Reduce cost of collection and resistance from taxpayers

Expanding our knowledge

Findings from other relevant studies

- Tax revenue and compliance increases with increased tech and enforcement
- Many results are from small-scale local experiments: Generalizable?

What remains?

- How to make technology and personnel complementary?
- How to tax new commodities (digital activities, digital currencies)?
- How to increase political support for taxation?

Jensen (2022):
Employment Structure and the Rise of the Modern Tax System

Q: How do modern tax systems rise?

Modern tax systems are characterized by increased direct taxes (income tax)

- Requires high collection capacities
- Changes in the structure of employment may explain increase in collection
  - Enforcement in employer-employee structure $>$ Self-employed
  - Existence of third-party information (employer, in case of income tax)

This paper uncovers new facts on tax capacity with microdata across countries and time

- Employee share increases over income distribution & further down the income distribution as country develops
- Exemption threshold gradually decreases in income distribution over development
- Threshold is located such that employee share on tax base remains constant and high

Data, definition, and calculation strategy

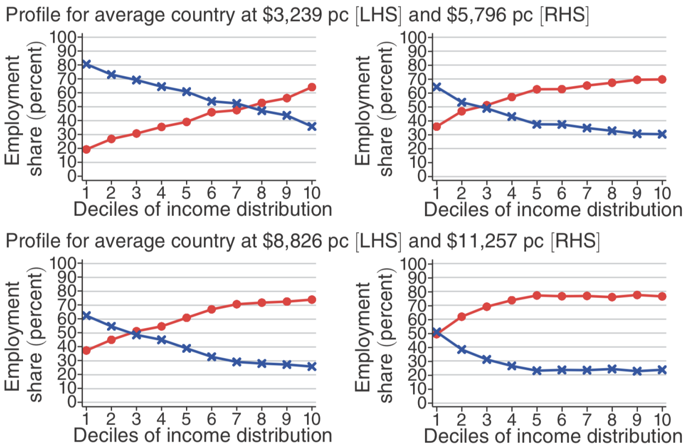
Data source and definitions

- Household surveys (100 countries)
  - Employment type of economically active population (EAP)
  - Information on taxable earnings (employment / self-employment / capital / other income)
  - For US (1870-2010) and Mexico (1960-2010), compile long run time series

Definition of key terms and strategies for calculation

- Employee: if work type generates third-party information usable in tax enforcement (Otherwise, self-employment)
  - Self-employed: casual laborers, own account workers, workers w/o contracts
  - Calculate share of employees in EAP across country's income distribution (in deciles)
- Threshold: Minimum level of gross income above which individuals have to pay taxes
  - From country tax reports of International Bureau of Fiscal Documentation
- Income tax base: % EAP whose gross income is above income tax exemption threshold

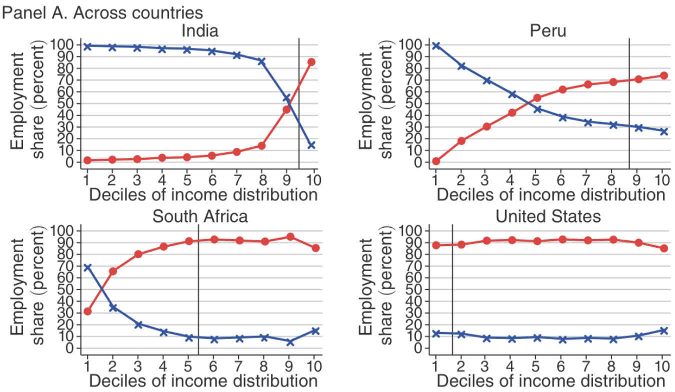
Fact1: Employee share ⇑ with income and development levels



Red: Employee, Blue: Self-employed

- Within country, employee share increase over income
- Across countries, employee share profile moves leftward
- Over transition, increase in employee share concentrated in deciles gradually further down

Fact2: Exemption threshold moves downward as country develops



Red: Employee, Blue: Self-employed, Vertical: Threshold

- At low development, threshold at top deciles
- With development, growth in employee share comoves with falling thresholds
- Tax base increases
- Documents frequent updating of thresholds (base increase not driven by nominal inflation)

Fact3: Threshold is located where employee share remains constant

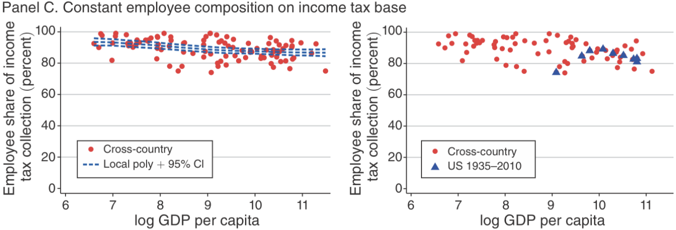


FIGURE 4. SIZE AND COMPOSITION OF INCOME TAX BASE OVER DEVELOPMENT

- Composition of tax base is constant and high in employee share
- Employee share: 85-95%
- Suggest that employee composition matters in deciding threshold
- % Self-employed $\Uparrow \rightarrow$ Enforcement costs $\Uparrow$ (high employee share desirable)

Takeaways from the stylized facts

Cost channels at play in expanding tax base

- Concerns with taxing and income segment with too high-self employed population
  - Employees have information trails, self-employed do not
- Compliance cost imposed on newly taxed self-employed may be high
  - Employees have third-party withholding
- Worsen between-group horizontal equity among employee vs self-employed
  - Self-employed have evasion opportunities $\rightarrow$ effective taxes paid < legal liabilities

Rise of modern taxation: Decrease in threshold (broad-based tax) happens when

- Increase in employee share at lower end of income distribution (Fact 1)
- This lowers costs of reform and expand tax base (Fact 2)
- While retaining high employee share, making tax administration manageable (Fact 3)

Implications for developing countries

Redistribution by direct taxes might be ill-suited for developing countries

- Developed countries extensively use third-party information
  - Evidenced by increase in employee share down the income distribution over time
- Developing countries cannot enforce taxes flexibly due to cost constraints
  - Higher share of self-employment (difficult to gather info)
- For such settings, indirect taxation could be the best option for redistribution

Remaining issues

- How would implementation of technology affect tax base long-run?
  - More use of third-party information $\rightarrow$ Rising 'employee' share?
  - Expanding tax base would then be less costly $\rightarrow$ tax collection, public goods $\Uparrow$
  - But what about adoption (existence of political will)?

Olsson, Baaz, and Martinsson (2020): Fiscal capacity in post-conflict states: Evidence from trade on Congo River [student presentation]

Course logistics

Balán, Bergeron, Tourek, Weigel (2022): Local Elites as State Capacity: How City Chiefs Use Local Information to Increase Tax Compliance in the DRC

Q: How can local information be used effectively to increase taxes?

How can fiscal capacities develop in presence of local & traditional elites

- On the one hand, can capture politics and civic society
- On the other, could be room to collaborate to improve governance
- Potential trade-offs
  - Enforcement backed by the state vs local knowledge and coordination?

This study investigates if local elites can contribute to statebuilding

- Comparison between state agents vs local elites
  - Local elites know more contextual knowledge but could lead to leakage in revenues
  - State agents have better enforcement capacity
- Based on field experiment randomizing type of tax collectors across neighborhoods
- Estimate changes in revenues and bribes, with tests for various mechanism

Taxation in Democratic Republic of Congo (DRC)

Setting of DRC

- Low-capacity state: tax/GDP ranks 188 out of 200 and \$0.30 per capita tax revenue
- Relies on property tax: First campaign to expand property tax in 2016
- Public goods and services are scarce and low-quality
  - Limited running water, electricity, unrepaired roads etc

The current study focuses on the second campaign (May-Dec 2018)

- Collectors trained by provincial ministry, separate for state and chiefs
- Involved property registration updates, providing liability information
- Followed by tax visits (revisit until taxes are paid)
- Collectors compensated based on houses registered and revenue collection
- Property tax rates are tiered flat, fixed fees (low: \$2 tax, 89% vs high: \$9 tax, 11%)
  - Equivalent to 0.32% of property value ($\simeq$ lower end of property tax rates in US)

Data and empirical strategy

RCT on 45000+ properties in 356 neighborhoods in Kasai-Central

- Randomized at neighborhood, treatment at past campaigns, experience of chiefs
- Central (state agents) vs Local (chiefs) vs State agents w/ local info vs state and local
- Pure control group: Declare their property values voluntarily
- Other aspects of collection equal
  - Property registration, assessment, tax liabilities, training, collection compensation
- Complemented with tax administrative data and household surveys

Estimation leverages random assignment of treatment (indiv i, neighbor j, strata k, time t)

$$y_{ijkt} = \beta_0 + \beta \text{Local}_{jkt} + \gamma X_{ijkt} + \alpha_k + \theta_t + e_{ijkt}$$

- For mechanisms, other treatment arms included as separate variable

Increase in compliance and revenues

TABLE 4—LOCAL VERSUS CENTRAL: COMPLIANCE AND REVENUES

	(1)	(2)	(3)	(4)	(5)
Panel A. Compliance					
Local	0.023	0.032	0.032	0.033	0.040
	(0.008)	(0.007)	(0.008)	(0.007)	(0.008)
Observations	28,872	27,764	213	27,764	23,803
Clusters	221	213		213	213
Central mean	0.068	0.063	0.065	0.063	0.073
Panel B. Revenues					
Local	57.627	79.640	81.830	68.855	81.991
	(25.688)	(22.856)	(38.595)	(20.560)	(23.562)
Observations	28,872	27,764	213	27,764	23,803
Clusters	221	213		213	213
Mean	192.891	182.236	210.134	182.236	208.568
Time fixed effects	No	Yes	Yes	Yes	Yes
House fixed effects	No	No	No	Yes	Yes
Stratum fixed effects	Yes	Yes	Yes	Yes	Yes
Exempt excluded	No	No	No	No	Yes

Accurate assessments & trust in government ↑, but bribes ↑

Table 5—Local versus Central: Mismanagement and Views of Government, Chiefs, and Taxes

	$\hat{\beta}$	SE	R^2	Observations	$\bar{x}_{Central}$
Panel A. Property assessments					
Assigned exemption	0.039	0.021	0.055	13,772	0.266
Incorrect exemption	−0.012	0.007	0.020	13,771	0.044
Assigned high band	0.030	0.021	0.230	27,764	0.114
Incorrect assignment	−0.013	0.006	0.041	27,764	0.031
Panel B. Bribes					
Paid bribe (midline)	−0.001	0.003	0.007	18,596	0.016
Gap self versus admin (midline)	0.016	0.009	0.018	14,309	0.077
Paid bribe (endline)	0.018	0.009	0.049	1,169	0.014
Other payments (endline)	0.031	0.014	0.041	2,407	0.094
Panel C. View of government					
View of government (index)	0.023	0.049	0.100	2,411	0.011
Trust in government	0.127	0.057	0.075	2,286	0.028
Responsiveness of government	−0.049	0.045	0.099	2,282	0
Performance of government	−0.060	0.052	0.060	2,179	−0.014
Integrity of government	0.043	0.047	0.058	2,313	0.016
Panel D. View of taxation					
Perceived tax compliance on avenue	0.100	0.055	0.073	1,851	0.026
Trust in tax ministry	0.085	0.061	0.073	2,259	0.025
Property tax morale	0.075	0.047	0.057	2,343	0.014
Fairness of property taxation	−0.004	0.053	0.046	2,407	0.003
Perception of enforcement	−0.019	0.058	0.070	2,379	0.015

Evidence of local information at play

TABLE 7—CENTRAL VERSUS CLI

	Compliance (1)	Revenues (2)	Visited (3)	Visits (4)	Compliance (5)	Compliance (6)
CLI	0.024	46.566	−0.016	−0.026	0.026	0.022
	(0.009)	(21.200)	(0.028)	(0.044)	(0.014)	(0.009)
Local						0.046
						(0.007)
Visit control	No	No	No	No	Yes	No
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
House fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Stratum fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	20,636	20,636	13,884	13,877	5,283	33,746
Clusters	165	165	163	163	161	267
Central mean	0.051	150.66	0.387	0.497	0.097	0.052
Test CLI = Local (p-value)						0.007

Discussion and takeaways

What explains the chiefs' advantage? Better targeting with information advantage

- Lower effort cost? Chiefs did not make more visits
- Target their visits? When state agents received info from chiefs, compliance $\Uparrow$
- Better persuasion? Neutralize info advantage (visit in linear route)$\Rightarrow$ Central = Local

Takeaways and remaining questions

- Should countries with low capacities rely on local elites?
  - Raise revenues, but also bribes
  - To shift to state, value of \$1 lost in bribes $>>$ \$1 gained in revenues
- For technology to increase taxes, need to surpass some state capacity threshold?
  - Speak to the complementary/substitutional relationship between tech and person

Henn, Mugaruka, Ortiz, Sánchez de la Sierra, Wu (2025): Monopoly of Taxation without a Monopoly of Violence: The Weak State's Trade-off from Taxation

Q: Tradeoffs between monopoly of taxation & monopoly of violence?

Weak states often fail to exert de facto control over their jurisdictions

- Non-state actors (rebels) take over, who sometimes provide stability over
- If these areas are economically valuable, state loses lot of fiscal revenue
- So is it beneficial for state to take formal authority over those lands?
    - Regain lost fiscal revenues
    - But could trigger inadvertent response from rebels (e.g. plundering)

This paper investigates the effect of state-building attempts to increase tax revenues

- Military powers to reclaim land and replace rebels
- Negotiate with rebels (often with concession to the rebels)
- Leverage subnational evidence in Democratic Republic of Congo (DRC)

State-building efforts to drive rebels out of DRC

State absence in Eastern DRC (and postcolonial Africa in general)

- Many parts of DRC (East) were under armed group control since 1998
- Most notable of these organizations is FDLR (Forces de Libération du Rwanda)

Military operation and peace treaty with rebels

- Largest peace treaty: Sun City peace treaty in 2004
  - Various armed groups integrated into Congolese army
  - Allowed government to regain half the country without triggering violence
  - Other peace agreements had smaller goals and lower credibility
- Largest military operation: Kimia II (March 2009)
  - Joint operation with UN, national army, and Rwanda
  - Targeted FDLR presiding and taxing over Basile Chiefdom region since 2005
  - Although defeated, FDLR launched theft operations from the nearby hilly forest region

Data and empirical strategy

Constructing the data

- Armed group presence: village-level data through interviews with former combatants
  - Perpetrators' affiliation, motivation (pilage/retaliate/etc), actions taken

Effects of Kimia II and Sun City Peace Treaty

- Kimia II: FDLR vs non-FDLR villages pre ('05-'09) and post ('10-'12) in Basile Chiefdom
- Peace treaty: Integrated villages vs others in the region before and after '02

$$Y_{i,t} = \alpha_i + \beta_t + \sum_{k=-4}^{k=3} \beta_k \text{Treat}_i \times 1[t = t_0 + k] + e_{i,t}$$

  - $t_0 = 2009$ for Kimia II effects, 2002 for peace treaty effects

Kimia II: FDLR increased violence afterwards

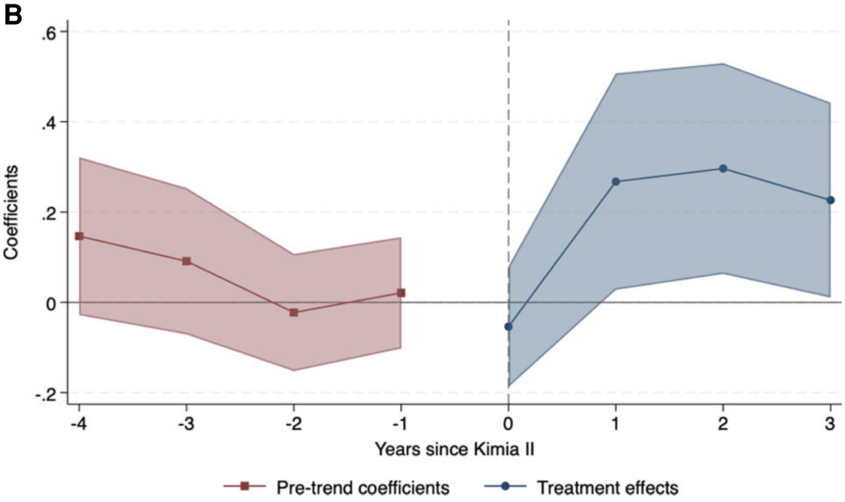


FIGURE 4

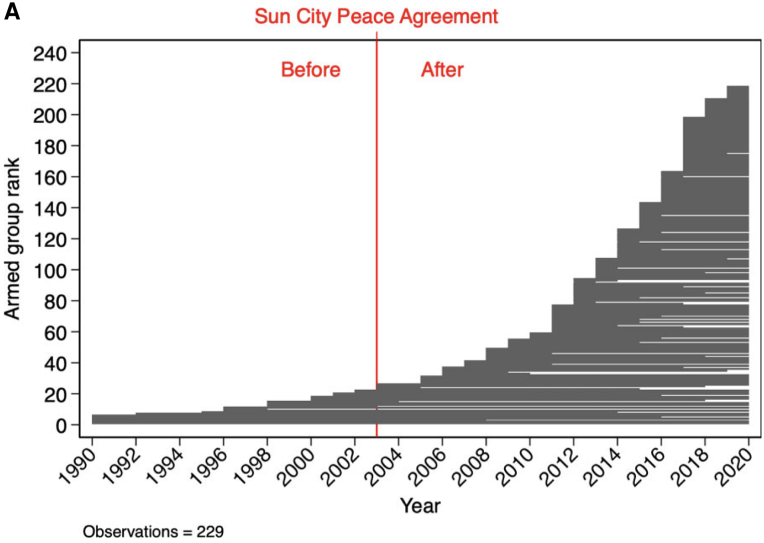
Kimia II campaign and FDLR violence, for FDLR state villages and the rest (A) Times Series and (B) Event Study

Kimia II: Driven by increase in pillages by FDLR

TABLE 1
Mechanism—purported motivation for the attacks

	Dependent variable: FDLR attack						
	Any (1)	Conquest (2)	Punishment (3)	Pillage (4)	Conquest (5)	Punishment (6)	Pillage (7)
$\text{FDLR}_i \times 1(t > 2009)$	0.240***	−0.001	0.043***	0.220***	−0.030	0.037***	0.210***
	(0.068)	(0.018)	(0.006)	(0.063)	(0.020)	(0.006)	(0.060)
Control FDLR Conquest	N	N	N	N	N	Y	Y
Control FDLR Punishment	N	N	N	N	Y	N	Y
Control FDLR Pillage	N	N	N	N	Y	Y	N
Observations	3,474	3,474	3,474	3,474	3,474	3,474	3,474
R^2	0.14	0.08	0.08	0.13	0.29	0.28	0.17
Mean Dep. Var.	0.07	0.02	0.02	0.04	0.02	0.02	0.04
Village clusters	193	193	193	193	193	193	193
Chiefdom-year Clusters	360	360	360	360	360	360	360
Conley (1999) *p*-value	0.00	0.97	0.00	0.00	0.12	0.00	0.00

Peace treaty: Rise of new armed groups since treaty



Takeaways and discussions

What (doesn't) explain the effects

- Actual violence increased (as opposed to *reported* violence)
- Violence by other actors do not increase following Kimia II
  - Thus when incentive to protect villages (for tax revenues) vanish, violence increase
- Treaty raised revenues, but limited commitment ability led to questioning of legitimacy

Takeaways and questions

- Kimia II: Monopolize violence (rebels out), but lost taxation due to ensuing violence
- Peace treaty: Expand tax revenues but no longer monopolize violence
- If faced with these trade-offs what should the states do?
  - Asserting force desirable if state has more capacity to prevent crime
  - Bargaining potentially desirable if states can commit to not negotiate with future rebels and have resource to integrate rebels

Ch, Shapiro, Steel, Vargas (2018): Endogenous Taxation in Ongoing Internal Conflict: The Case of Colombia [student presentation]

Takeaways and remaining questions

Trajectory of fiscal capacity development

- More broad-based taxation (income tax) aided by investments in capacity
- Many barriers to investment in developing countries
  - Informal economies that distort taxation decisions
  - High enforcement cost due to lack of technology, third-party info, and personnel
  - Establishing legitimacy difficult in face of internal conflicts
- Utilization of local knowledge possible: But need to be wary of local capture/bribes

Remaining questions

- Other input in state capacity: Personnel to execute tasks
  - Responsible for collecting taxes, delivering public services
  - How do you recruit talented, motivated bureaucrats? How do you retain them?
  - How do you incentivize bureaucrats and monitor them to prevent deviant actions?